

DeepSense 6G Scenario 36: Technical Dataset Deep-Dive

Exhaustive Feature Schemas, 24,799 Sample Multi-Modal Time-Series, and Zero-Leakage Splitting

1. Multi-Modal Data Architecture & Sensor Hardware

Scenario 36 provides real-world multi-modal sensing and communication data collected across 124 continuous vehicle drives in urban and highway environments. The receiver vehicle (Unit 1, Ford Transit) is equipped with four 64-element phased arrays operating at 60 GHz, forward-looking ZED2 stereo cameras, and RTK-GPS. The transmitter vehicle (Unit 2, Lincoln MKZ) transmits an omnidirectional 60 GHz beacon signal with synchronized GPS telemetry.

2. 12 Engineered Spatial Kinematics & Mathematical Formulations

Feature Name	Symbol & Formula	Physical Range	Engineering & Physical Significance
rel_lat	$\Delta_{lat} = Lat_{Unit2} - Lat_{Unit1}$	[-0.002, +0.002] deg	North-South spatial displacement between vehicles
rel_lon	$\Delta_{lon} = Lon_{Unit2} - Lon_{Unit1}$	[-0.002, +0.002] deg	East-West spatial displacement between vehicles
distance	$d = \sqrt{\Delta_{lat}^2 + \Delta_{lon}^2} * 111,000$	[2.1, 229.4] meters	Physical Line-of-Sight Euclidean distance (LOS path loss indicator)
bearing	$\theta = \arctan2(\Delta_{lon}, \Delta_{lat})$	[-pi, +pi] radians	Geometric azimuth angle from Unit 1 antenna array to Unit 2
sin_bearing	$\sin(\theta)$	[-1.0, +1.0]	Trigonometric projection eliminating the 0 to 2pi wraparound discontinuity
cos_bearing	$\cos(\theta)$	[-1.0, +1.0]	Orthogonal trigonometric projection for smooth neural network gradient flow
bearing_rate	$d(\theta)/dt = \text{grad}(\theta)$	[-1.8, +1.8] rad/s	Angular velocity (how fast the transmitter is turning across the field of view)
distance_rate	$d(d)/dt = \text{grad}(d)$	[-25.0, +25.0] m/s	Relative range closing speed (Doppler shift and path loss rate)
unit1_speed	v_1 (instantaneous speed)	[0.0, 32.5] m/s	Speed of the receiver vehicle (Ford Transit)
unit1_heading	phi_1 (compass heading)	[0.0, 360.0] deg	True heading direction of the receiver vehicle
unit2_speed	v_2 (instantaneous speed)	[0.0, 34.2] m/s	Speed of the transmitter vehicle (Lincoln MKZ)
unit2_heading	phi_2 (compass heading)	[0.0, 360.0] deg	True heading direction of the transmitter vehicle

3. Visual Modality (ZED2 RGB Camera) & RF Power Profiles (256 Beams)

ZED2 Camera Integration: Forward-looking RGB frames (`unit1_rgb5` and `unit1_rgb6`) capture visual road topology, leading vehicle silhouette, and potential visual obstructions. Images are resized to 96x96x3, normalized via ImageNet statistics, and processed through a fine-tuned ResNet-18 visual encoder.

256-Element Beam Power Profiles: Unit 1 houses 4 planar phased-array panels, each producing 64 directional narrow beams ($4 \times 64 = 256$ beams). The ground-truth power measurements in `scenario36.p` are pre-stored in linear milliwatt (mW) values and converted to logarithmic scale: $P_{\text{dBm}} = 10 \log_{10}(P_{\text{linear}}) + 10^{-12}$. The index maximizing received power is the target label `unit1_overall-beam` (0 to 255).

4. Drive-Block Splitting & The 0% Spatial Cluster Leakage Bug

The Naive Chronological Split Hazard: A standard sequential chronological split (e.g. first 70% samples for training, last 15% for testing) causes severe data failure: all test samples end up localized in geographic regions never seen during training (0% spatial overlap). Furthermore, 3-second sliding windows overlap by 29 time steps between adjacent rows, causing artificial memorization.

Our Zero-Leakage Drive-Block Solution: We group all 24,799 samples strictly by continuous driving trajectory runs (`seq_index`, 124 distinct drives). Entire driving runs are randomly partitioned into Train (70%, 17,199 samples), Calibration (15%, 3,600 samples), and Test (15%, 4,000 samples), guaranteeing true generalization on unseen trajectories.